

Circularly polarized Raman scattering for optical spin detection in silicon metasurfaces

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Abstract— We use polarization-resolved Raman scattering to probe optical spin in silicon metasurfaces under circular excitation. By tuning disk size and wavelength, we observe strong Raman enhancement with far field polarization losses. Simulations show that quadrupolar resonances spatially varying near fields that depolarize the emitted Raman signal.

Keywords— high-index dielectric metasurfaces, Raman scattering, circular polarization, Mie resonances.

Nanostructured dielectric metasurfaces provide a powerful platform for tailoring polarization in light-matter interactions for advanced photonic functionalities [1]. However, experimentally accessing the polarization state of optical evanescent fields remains challenging. Raman scattering in silicon, with a well-known polarization response in bulk [2], is an attractive way to address this issue. Despite Raman signal enhancements have been widely demonstrated in nanophotonic structures with strong electromagnetic confinements, its polarization response in such situations remains unexplored [3].

Here, we experimentally explore Silicon on Sapphire disk metasurfaces with polarization-resolved Raman spectroscopy (Fig. 1a). Using circularly polarized light for illumination, we relate the measured Raman enhancement and its Degree of Circular Polarization (DOCP) to Mie resonances in the metasurface, and we link these to the internal field optical spin with numerical simulations.

Our experimental results show, in different metasurfaces:

- Strong Raman enhancement with high DOCP.
- Strong Raman enhancement with strong DOCP losses.

Simulations, together with multipolar analysis, show:

- A good agreement between simulation and experiment.
- DOCP drops connected to angled collection in moderate NA objectives.
- Intrinsic loss of DOCP related to quadrupolar resonances in the metasurface (Fig. 1b).

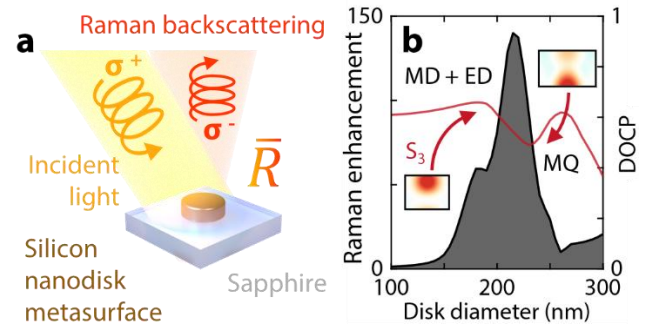


Fig 1. a) Concept art of the studied metasurface. b) Correlation of Raman enhancing resonances with DOCP in simulations.

In conclusion, quadrupolar modes produce complex near-field distributions with spatially varying optical spin, leading to polarization loss in the Raman emission. Polarization-resolved Raman scattering is a useful probe of near-field polarization and optical spin in dielectric nanophotonic systems.

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